

Christopher G. Goemans

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EDUCATION

2006	Ph.D. Economics	University of Colorado
2000	M.A. Economics	University of Colorado
1997	B.A. Economics	University of Maine

ACADEMIC POSITIONS

Summer 2021 - Present	Professor	Department of Agricultural and Resource Economics	Colorado State University
Summer 2013 - Spring 2021	Associate Professor	Department of Agricultural and Resource Economics	Colorado State University
Fall 2007-Spring 2013	Assistant Professor	Department of Agricultural and Resource Economics	Colorado State University
Summer 2008	Visiting Scholar	Institute for the Study of Competition and Regulation	Victoria University of Wellington
Summer 2006-Summer 2007	Postdoctoral Research Associate	Cooperative Institute for Research in Environmental Science	University of Colorado
Fall 2003 to Spring 2004	Adjunct Instructor	Department of Economics	University of Colorado at Denver

OTHER RELEVANT EMPLOYMENT EXPERIENCE

Fall 2003-Fall 2005	Senior Associate	Harvey Economics	Denver, Colorado
Fall 2002	Assistant Contractor	Brown and Caldwell	Denver, Colorado

HONORS AND AWARDS

2018 Outstanding Published Research Award (awarded for: "Dilemmas, coordination and defection: How uncertain tipping points induce common pool resource destruction." Games and Economic Behavior 104 (2017): 760-774.)

2014 North American College and Teachers of Agriculture (NACTA)/Charles N. Shepardson Meritorious Teaching Award

2012 CSU College of Agricultural Sciences Team Research Award (Goemans et al.)

2010 Charles N. Shepardson Faculty Teaching Award

2010 (nominee) CSU Alumni Association's Best Teacher

Alpha Gamma Rho's Honored Professor (fall 2008, fall 2009, spring 2009, fall 2010, spring 2012)

Recognized by the athletic department at the 2009 Ram-Scholar Athlete brunch as "one of our student athletes' most influential faculty members"

2007 University Council of Water Resources Honorable Mention Ph.D. Dissertation Award in Water Policy & Socioeconomics

PUBLISHED WORKS

* indicates co-author was a graduate student under my supervision when manuscript was developed

REFEREED ARTICLES

41. Rouhi Rad, Mani, Jordan F. Suter, Dale T. Manning, and Christopher Goemans. "Abatement subsidies for groundwater conservation." *American Journal of Agricultural Economics* (2025).

40. *Muriqi, Diellza, Jude Bayham, Christopher Goemans, Dale T. Manning, and Jordan F. Suter. "Inland flooding in the United States: a review of research related to risk management, economic behavior, and data resources." *Environmental Research Letters* (2025).

39. Sloggy, Matthew R., Aaron Hrozencik, Dale T. Manning, Chris G. Goemans, and Roger L. Claassen. "Insurance and Extraction Incentives in a Common Pool Resource: Evidence from Groundwater Use in the High Plains." *Journal of Environmental Economics and Management* (2025): 103125.

38. *Gharib, Ahmed A., Mazdak Arabi, Christopher Goemans, Dale T. Manning, and Alexander Maas. "Integrated water management under different water rights institutions and population patterns: Methodology and application." *Water Resources Research* 60, no. 11 (2024): e2024WR037196.

37. Maas, Alexander, Roshan Puri, and Christopher Goemans. "A review of residential water conservation policies and attempts to measure their effectiveness." *PLOS Water* 3, no. 8 (2024): e0000278.

36. *Gharib, Ahmed A., *Joey Blumberg, Dale T. Manning, Christopher Goemans, and Mazdak Arabi. "Assessment of vulnerability to water shortage in semi-arid river basins: The value of demand reduction and storage capacity." *Science of The Total Environment* (2023): 161964.

35. *Lurbé, Salvador, Jesse Burkhardt, Chris Goemans, Dale Manning, and Liesel Hans. "Further evidence on social comparison and residential water use." *Water Resources and Economics* 41 (2023): 100214.

34. Filippelli, Steven K., Matthew R. Sloggy, Jody C. Vogeler, Dale T. Manning, Christopher Goemans, and Gabriel B. Senay. "Remote sensing of field-scale irrigation withdrawals in the central Ogallala aquifer region." *Agricultural Water Management* 271 (2022): 107764.

33. *Blumberg, Joey, Chris Goemans, and Dale Manning. *Producer Beliefs and Conservation: The Impact of Perceived Water Scarcity on Irrigation Technology Adoption*. No. w30080. National Bureau of Economic Research (2022).

32. *Hrozencik, R. Aaron, Dale T. Manning, Jordan F. Suter, and Christopher Goemans. "Impacts of Block-Rate Energy Pricing on Groundwater Demand in Irrigated Agriculture." *American Journal of Agricultural Economics* 104, no. 1 (2022): 404-427.
31. Suter, Jordan F., Mani Rouhi Rad, Dale T. Manning, Chris Goemans, and Matthew R. Sanderson. "Depletion, climate, and the incremental value of groundwater." *Resource and Energy Economics* 63 (2021): 101143.
30. Rouhi Rad, Mani, Dale T. Manning, Jordan F. Suter, and Christopher Goemans. "Policy Leakage or Policy Benefit? Spatial Spillovers from Conservation Policies in Common Property Resources." *Journal of the Association of Environmental and Resource Economists* 8, no. 5 (2021): 923-953.
29. Sloggy, Matthew R., Jordan F. Suter, Mani Rouhi Rad, Dale T. Manning, and Chris Goemans. "Changing climate, changing minds? The effects of natural disasters on public perceptions of climate change." *Climatic Change* 168 (2021): 1-26.
28. Neale, Michael R., Sybil Sharvelle, Mazdak Arabi, Andre Dozier, and Chris Goemans. "Assessing tradeoffs of strategies for urban water conservation and fit for purpose water." *Journal of Hydrology X* 8 (2020): 100059.
27. Manning, Dale, Mani Rouhi Rad, Jordan Suter, Christopher Goemans, Zaichen Xiang, and Ryan Bailey. "Non-market Valuation in Integrated Assessment Modeling: The Benefits of Water Right Retirement." *Journal of Environmental Economics and Management* (2020): 102341.
26. *Janusch, Nicholas, Stephan Kroll, Christopher Goemans, Todd L. Cherry, and Steffen Kallbekken. "Learning to accept welfare-enhancing policies: an experimental investigation of congestion pricing." *Experimental Economics* (2020): 1-28.
25. *Maas, Alexander, Christopher Goemans, Dale T. Manning, Jesse Burkhardt, and Mazdak Arabi. "Complements of the house: Estimating demand-side linkages between residential water and electricity." *Water Resources and Economics* 29 (2020): 100140.
24. Burkhardt, Jesse, and Chris Goemans. "The short-run effects of marijuana dispensary openings on local crime." *The Annals of Regional Science* 63, no. 1 (2019): 163-189.
23. Zhong, Hua, Michael H. Taylor, Kimberly S. Rollins, Dale T. Manning, and Christopher G. Goemans. "Who pays for water scarcity? Evaluating the welfare implications of water infrastructure investments for cities." *The Annals of Regional Science* 63, no. 3 (2019): 559-600.
22. *Shepler, Ryan, Jordan F. Suter, Dale T. Manning, and Chris Goemans. "Private Actions and Preferences for Coordinated Groundwater Conservation in Colorado's Republican River Basin." *Journal of the American Water Resources Association* 55, no. 3 (2019): 657-669.
21. *Flyr, Matthew, Jesse Burkhardt, Christopher Goemans, Liesel Hans, Abbye Neel, and Alexander Maas. "Modeling Commercial Demand for Water: Exploring Alternative Prices, Instrumental Variables, and Heterogeneity." *Land Economics* 95, no. 2 (2019): 211-224.
20. *Stone, Janine, Christopher Goemans, and Marco Costanigro. "Variation in Water Demand Responsiveness to Utility Policies and Weather: A Latent-Class Model." *Water Economics and Policy (WEP)* 6, no. 01 (2019): 1-33.
19. Hoag, Dana LK, Chris Goemans, and *Anthony Orlando. "Sustainable policies that align irrigation and water quality." In *Western Economics Forum*, vol. 16, no. 1837-2018-2992, pp. 54-61. 2018.
18. *Stone, Janine, Marco Costanigro, and Christopher Goemans. "Public opinion on Colorado Water rights transfers: Are policy preferences consistent with concerns over impacts?." *Journal of agricultural and resource economics* 43, no. 1835-2018-3858 (2018): 403-422.

17. *Hrozencik, Robert A., Dale T. Manning, Jordan F. Suter, Christopher Goemans, and Ryan T. Bailey. "The heterogeneous impacts of groundwater management policies in the Republican River Basin of Colorado." *Water Resources Research* 53, no. 12 (2017): 10757-10778.
16. Manning, Dale T., Christopher Goemans, and *Alexander Maas. "Producer responses to surface water availability and implications for climate change adaptation." *Land Economics* 93, no. 4 (2017): 631-653.
15. *Dozier, André Q., Mazdak Arabi, *Benjamin C. Wostoupal, Christopher G. Goemans, *Yao Zhang, and Keith Paustian. "Declining agricultural production in rapidly urbanizing semi-arid regions: policy tradeoffs and sustainability indicators." *Environmental Research Letters* 12, no. 8 (2017): 085005.
14. *Maas, Alexander, Christopher Goemans, Dale Manning, Stephan Kroll, and Thomas Brown. "Dilemmas, coordination and defection: How uncertain tipping points induce common pool resource destruction." *Games and Economic Behavior* 104 (2017): 760-774.
13. *Maas, Alexander, Christopher Goemans, Dale Manning, Stephan Kroll, Mazdak Arabi, and Mariana Rodriguez-McGoffin. "Evaluating the effect of conservation motivations on residential water demand." *Journal of Environmental Management* 196 (2017): 394-401.
12. *Maas, Alexander, Andre Dozier, Dale T. Manning, and Christopher Goemans. "Water storage in a changing environment: the impact of allocation institutions on value." *Water Resources Research* 53, no. 1 (2017): 672-687.
11. Strong, Aaron, and Chris Goemans. "The impact of real-time quantity information on residential water demand." *Water Resources and Economics* 10 (2015): 1-13.
10. *Swanepoel, G. D., Joleen Hadrich, and Christopher Goemans. "Estimating the contribution of groundwater irrigation to farmland values in Phillips County, Colorado." *Journal of the ASFMRA (American Society of Farm Managers and Rural Appraisers)* 2015, no. 387-2016-23044 (2015): 166-179.
9. Strong, Aaron, and Chris Goemans. "Quantity uncertainty and demand: The case of water smart reader ownership." *The BE Journal of Economic Analysis & Policy* 14, no. 3 (2014): 669-694.
8. *Bauman, Allison, Christopher Goemans, James Pritchett, and Dawn Thilmany McFadden. "Estimating the economic and social impacts from the drought in Southern Colorado." *Journal of Contemporary Water Research & Education* 151, no. 1 (2013): 61-69.
7. Goemans, Christopher, Marco Costanigro, and *Janine Stone. "The interaction of water restriction and pricing policies: econometric, managerial and distributional implications." *Journal of Natural Resources Policy Research* 4, no. 1 (2012): 61-77.
6. Costanigro, Marco, Jill J. McCluskey, and Christopher Goemans. "The economics of nested names: name specificity, reputations, and price premia." *American Journal of Agricultural Economics* 92, no. 5 (2010): 1339-1350.
5. *Thorvaldson, Jennifer, James Pritchett, and Christopher Goemans. "Western households' water knowledge, preferences, and willingness to pay." *Canadian Journal of Agricultural Economics/Revue canadienne d'agroeconomie* 58, no. 4 (2010): 497-514.
4. Kenney, Douglas S., Christopher Goemans, Roberta Klein, Jessica Lowrey, and Kevin Reidy. "Residential water demand management: lessons from Aurora, Colorado 1." *JAWRA Journal of the American Water Resources Association* 44, no. 1 (2008): 192-207.
3. Howe, Charles W., and Christopher Goemans. "The simple analytics of demand hardening." *Journal - American Water Works Association* 99, no. 10 (2007): 24-25.

2. Howe, Charles W., and Christopher Goemans. "Water Transfers and Their Impacts: Lessons from three Colorado Water Markets." *Journal of the American Water Resources Association* 39.5 (2003): 1055-1065.
1. Howe, Charles W., and Chris Goemans. "Effectiveness of water rate increases following watering restrictions." *Journal of the American Water Works Association* 94, no. 10 (2002): 28-32.

BOOKS/BOOK CHAPTERS

3. *Blumberg, Joey, Goemans, Christopher and Manning, Dale. "5. Perceived Water Scarcity and Irrigation Technology Adoption" In *American Agriculture, Water Resources, and Climate Change* edited by Gary D. Libecap and Ariel Dinar, 173-202. Chicago: University of Chicago Press, 2024. <https://doi.org/10.7208/chicago/9780226830629-007>
2. Goemans, Christopher, *Alexander Maas. 2016. "Northern Water Conservancy District." in Ronald Griffin's *Water Resource Economics: The Analysis of Scarcity, Policy, and Projects.*, second ed., The MIT Press.
1. Goemans, Christopher, and James Pritchett. 2014. "Western water markets: Effectiveness and efficiency." in *Water markets for the 21st century*, pp. 305-330. Springer, Dordrecht.

REFEREED PROCEEDINGS (FULL ARTICLE PUBLISHED)

1. *Thorvaldson, Jennifer, James Pritchett, Allen Bright, A. Shortsleeve, and Christopher Goemans, "Water Use in the Western U. S. Household Knowledge, Priorities and Willingness-to-Pay." USCID Fifth International Conference: Irrigation and Drainage for Food, Energy and the Environment. Salt Lake City, UT. November 5, 2009.

NON-REFEREED ARTICLES

20. Blumberg, Joey, Chris Goemans, and Dale Manning, "Changing Climate, Changing Institutions: Implications of Drought and Litigation for Colorado Agriculture." *Colorado Water*, Volume 38, Issue 2, 2021.
19. Falkowski, Michael, Dale Manning, Jillian LaRoe, Drew Bennett, Keith Carlisle, *Kevin Crofton, and Chris Goemans. "Quantifying the Scope and Regional Economic Impact of Permanent Agricultural Dry-Up Due to Rural to Urban Water Transfers." *Colorado Water*, Volume 35, Issue 5, 2018.
18. Guerrero, Bridget, Bill Golden, Karina Schoengold, Jordan Suter, Art Stoecker, Chris Goemans, and Dale Manning. "Groundwater Laws Across the Ogallala Aquifer Region." *Colorado Water*, Volume 34, Issue 6, 2017.
17. *Maas, Alex, *Andre Dozier, Dale Manning, and Christopher Goemans. "The Value of Stored Water and Trading in the West: Lessons from the Colorado Big-Thompson." *Colorado Water*, Volume 32, Issue 1, 2015.
16. *Quay, Brian, Christopher Goemans, James Pritchett. "Developing a Composite Index for Colorado's Irrigated Agriculture and Wellbeing." *Colorado Water*, Volume 31, Issue 3, 2014.
15. *Serbina, Lariss and Christopher Goemans. "Describing and Quantifying Profit Risk Producers Face when Adopting Water Conserving Cropping Systems." *Colorado Water*, Volume 30, Issue 3, 2013
14. Goemans, Christopher, James Pritchett, and *Nelson Ron. "2012 drought in Colorado: estimates of foregone revenues, indirect and induced economic activity for the crop sector." *Production and farm management report* (Colorado State University. Dept. of Agricultural and Resource Economics); PFMR 13-03 (2013).

13. Goemans, Christopher, James Pritchett, and *Ron Nelson. "Adaptations to drought: evidence from ag producer survey." Production and farm management report (Colorado State University. Dept. of Agricultural and Resource Economics); PFMR 13-01. (2013).
12. *Nelson, Ron, Christopher Goemans, and James Pritchett. "Farmer resiliency under drought conditions." Production and farm management report (Colorado State University. Dept. of Agricultural and Resource Economics); PFMR 13-02 (2013).
11. *Smith, Meagan., Mazdak Arabi, and Christopher Goemans. "Quantifying the Relationship between Irrigation Activities and Size of Wetlands in a Northern Colorado Watershed." *Colorado Water*, Volume 29, Issue 3, 2012.
10. Pritchett, James, Christopher Goemans, and *Jennifer Thorvaldson. "Water as a Crop: Are South Platte Farmers Willing to Participate in Innovative Leasing Arrangements?" *Colorado Water*, Volume 28, Issue 5, 2012.
9. Hurd, Brian, Christopher Goemans, George Frisvold, and *Janine Stone. "Estimated Impacts of Climate Change Legislation on New Mexico Agriculture." *NM State AES Bulletin 801*, 2011.
8. *Thomas, Cathie, Goemans, Christopher, and Craig Bond. "The Costs and Benefits of Preventative Management for Zebra and Quagga Mussels." *Colorado Water*, Volume 27, Issue 5, 2010.
7. *Stone, Janine and Christopher Goemans. "The Complexities of Conservation Research Needed to Incorporate Conservation Savings Into Utility Water Supply Planning." *Colorado Water*, Volume 27, Issue 4, 2010.
6. Goemans, Christopher, Stephan Kroll, and Kelly DiNatale. "Alternatives to Permanent Water Transfers Using the Farmers Reservoir and Irrigation Company (FRICO) System." *Colorado Water*, Volume 27, Issue 1, 2010.
5. Goemans, Christopher and Stephan Kroll. 2010. "Water Market Alternatives to Permanent Transfers." *CSU College of Ag Newsletter*, Fall 2010.
4. *Kanable, Cassie, and Christopher Goemans. "Understanding Weather's Impact on Household Water Demand." *Colorado Water*, Volume 25, Issue 5, 2009.
3. Kenney, Douglas, Roberta Klein, and Christopher Goemans. "Preliminary Findings from Western Water Assessment's 'Water Rights and Climate Change' Project." *Intermountain West Climate Summary*, Volume 5, Issue 2, 2009.
2. Kenney, Douglas, Christopher Goemans, Roberta Klein, Jessica Lowrey, and Kevin Reidy. "Residential Water Demand Management in Aurora: Learning from the Drought Crisis." *Colorado Water*, February/March Issue, 2007.
1. Goemans, Christopher. "South Platte Regional Assessment Tool (SPRAT)." *Intermountain West Climate Summary*, April Issue, 2006.

AGENCY REPORTS

8. Dale Manning, Jesse Burkhardt, Chris Goemans, and Alex Mass. 2021. “An Analysis of the Impact of Drought on Agriculture, Local Economies, Public Health, and Crime Across the Western United States”, NIDIS Project Report.
7. Manning, Dale, *Aaron Hrozencik, Chris Goemans, and Jordan Suter. 2016. “An Economic Analysis of Management Strategies to Conserve Groundwater in the Republican River Basin of Colorado”, CWCB Project Report.
6. *Gunter, Allie, Christopher Goemans, James Pritchett, and Dawn Thilmany. 2012. “The Economic Impact of the 2011 Drought on Southern Colorado: A combined input-output and EDMP analysis.” Colorado Water Conservation Board Project Report.
5. *Nelson, Ron, James Pritchett and Christopher Goemans. 2012. “Survey Summary: Farm and Ranch Managers’ Responses to the 2011 Drought.” Colorado Water Conservation Board Project Report.
4. DiNatale, K, A. Hickman, C. Goemans, S. Kroll, H. Thompson, and B. Dereume. 2012. “Water Partnerships: An Evaluation of Alternative Agricultural Water Transfer Methods in the South Platte Basin.” Colorado Water Conservation Board Project Report.
3. Hurd, Brian., C. Goemans, George Frisvold, and *Janine Stone. 2012. “Impacts of Climate Change Legislation on Agriculture in the Rocky Mountain States: Arizona, Colorado, and New Mexico.”, American Farmland Trust Project Report.
2. Kenney, Douglas, Roberta Klein, Christopher Goemans, Christina Alvord, and Janine Shapiro. 2008. “The Impact of Earlier Spring Snowmelt on Water Rights and Administration: A preliminary Overview of Issues and Circumstances in the Western States.”, Western Water Assessment Project Report.
1. Klein, Roberta, Douglas Kenney, Jessica Lowrey, and Christopher Goemans. 2008. “Factors Influencing Residential Water Demand: A Review of the Literature”, Western Water Assessment Project Report.

CONTRACTS & GRANTS

EXTERNALLY-FUNDED PROJECTS AS PI

- (2019-2020) Assessing the Impacts of Drought on Ecosystem Service-Based Sectors, Criminal Activity, and Health Outcomes in Colorado, Colorado Water Conservation Board, **\$32,983** (CoPI: Dale Manning)
- (2019-2020) Assessing the Impacts of Drought on Ecosystem Service-based Sectors, Criminal Activity, and Health Outcomes in Arizona, Colorado, New Mexico, and Utah, DOC-NOAA, **\$74,006** (CoPIs: Dale Manning and Jesse Burkhardt)
- (2018-2025) Evaluating Alternative Water Institution Performance in Snow-Dominated Basins: Are Food Productions Systems at Risk from Changing Snow Water Availability?, USDA-NIFA, **\$565,662** (represents Dept. of Ag and Res Econ portion of project, total project budget: \$5,100,000; CoPI Dale Manning)
- (2013-2018) General Equilibrium Model of Ecosystem Services, USDA-USFS-Rocky Mountain Research Station, **\$80,166**
- (2012-2013) Colorado Household Preferences in Meeting Future Water Challenges, Colorado Water Conservation Board, **\$50,000** (includes James Pritchett Co-PI)

(2012-2013) Estimating the Short and Long-term Economic and Social Impacts Associated with the Current Drought in Southern Colorado: An EDM Analysis, Colorado Water Conservation Board, **\$35,000**

(2012-2013) Estimating the Short and Long-term Economic and Social Impacts Associated with the Current Drought in Southern Colorado: Survey of Producers, Colorado Department of Agriculture, **\$25,000**

(2011-2016) The Short & Long-term Impacts of Drought on the Structure of Regional Economics: Investigating the Farm Supply Chain, DOI-USGS, **\$5,000**

(2011-2016) Quantifying Risks Producers Face When Entering Agricultural Water Lease Contracts, DOI-USGS, **\$5,000**

(2010) Feasibility Study to Assess the Potential of Urban Water Conservation to Meet Colorado's Future Water Supply Needs, Colorado Water Conservation Board, **\$40,000**

(2009-2010) Estimating the Cost Effectiveness of Water Conservation Programs, Jointly funded by the Colorado Water Institute and Colorado Water Conservation Board, **\$85,000**

(2009-2010) Alternative Water Transfers in the South Platte Basin using the Farmers Reservoir and Irrigation Company System, Farmers Reservoir and Irrigation Company, Colorado Water Conservation Board, \$58,000 (represents Dept. of Ag and Res Econ portion of project, total project budget value **\$250,000** (includes Stephan Kroll (CoPI))

(2004) Evaluating the Potential for Agricultural Water Transfers: Satisfying Municipal Water Demands During the Next 40 Years, NOAA Regional Integrated Sciences and Assessments (Western Water Assessment), **\$10,000**

(2004) Usefulness of Climate Forecasts in Water Management in the South Platte", NOAA Regional Integrated Sciences and Assessments (Western Water Assessment), **\$6,500**

EXTERNALLY-FUNDED PROJECTS AS CO-PI

(2024-2026)) "Inland Flooding and Evacuation: Individual Decisions and Economic Impacts", NOAA, \$500,000 (Jordan Suter (PI))

(2020-2025) "Asset mapping to understand opportunities for agricultural economic development that reduce water consumption", Colorado Water Conservation Board, \$163,824 (Becca Jablonski (PI))

(2014-2017) Economic Analysis & Design of Policies to Reduce Colorado's Groundwater Use in the Northern High Plains Ground Water Basin, Colorado Water Conservation Board, **\$159, 882** (includes Dale Manning (PI) and Jordan Suter (CoPI))

(2015-2018) Rural to Urban Water Transfers, Climate Change, & the Future of Rural Agricultural Economies in the Semi-Arid West: A Comparative Regional Analysis, USDA-NIFA, **\$243,851** (represents Dept. of Ag and Res Econ portion of project, total project budget: \$498,394; includes Dale Manning (PI) and Harvey Cutler (CoPI))

(2018-2021) Crop Insurance and Groundwater Consumption in the Ogallala Aquifer Region, UDS-NIFA, **\$499,785** (includes Dale Manning (PI) and Michael Falkowski (CoPI))

(2013-2014) Developing Metrics for Colorado Agriculture's Production & Efficiency with Water Resources, Colorado Water Conservation Board, **\$43,131** (includes James Pritchett (PI))

(2015-2017) Evaluating the Impact of Economic Research Results on Attitudes Related to Groundwater Management Policies: A Randomized Control Trial, Johns Hopkins University, **\$61,065** (includes Jordan Suter (PI) and Dale Manning (CoPI))

(2012-2013) Estimating the Short and Long-term Economic and Social Impacts Associated with the Current Drought in Southern Colorado (Phase 2), Colorado Water Conservation Board, **\$50,000** (includes James Pritchett (PI)).

(2009-2010) Adaptive Management of Zebra and Quagga Mussels in Colorado, Jointly funded by the Colorado Water Institute and Colorado Water Conservation Board, **\$85,000** (includes Craig Bond (PI))

(2006-2007) Residential Water Demand Management: Lessons from Aurora, Colorado, NOAA Regional Integrated Sciences and Assessments (Western Water Assessment), **\$60,000** (includes Douglas Kenney (PI))

(2004-2005) Assessing the Impacts of Climate Variability, Population Growth, and Changing Infrastructure on Future Water Supplies and Demands, NOAA Regional Integrated Sciences and Assessments (Western Water Assessment), **\$85,000** (includes Douglas Kenney (PI))

EXTERNALLY-FUNDED PROJECTS AS INVESTIGATOR OR ROLE OTHER THAN PI OR CO-PI

(2014-present)

(2018-2023) Interdisciplinary Training, Education and Research for Food-Energy-Water Systems (InTERFEWS) in Semi-Arid Regions, NSF, **\$2,994,781** (key person, includes Sybil Sharvelle (PI))

(2016-2020) Sustaining Agriculture through Adaptive Management to Preserve the Ogallala Aquifer under a Changing Climate, **\$404,725** (represents Dept. of Ag and Res Econ portion of project, total project budget: \$9,748,443; key person, includes Dale Manning and Jordan Suter within Dept.)

(2011-2013) Lower South Platte Irrigation Research And Demonstration Project, Parker Water and Sanitation, **\$384,000** (key person, includes Neil Hansen (PI))

INTERNALLY-FUNDED AWARDS

(2008) Development of a Research Working Group for Sustainable Urban Water System, School of Global Environmental Sustainability, **\$15,000** (CoPI, includes Larry Roesner (PI) Sybil Sharvelle (CoPI))

PAPERS PRESENTED

^indicates invited

*indicates graduate student

73. * “The value of demand reduction and storage capacity in alleviating predicted water shortage in semi-arid river basins.”, 2023 UCOWR/NIWR Annual Water Resources Conference, Fort Collins, Colorado, June 14, 2023 (presenter: Ahmed Gharib, included: Mazdak Arabi and Dale Manning)

72. ^ “Water Resource Allocation in the West”, CSU College of Agriculture Seminar Series, Fort Collins, Colorado, April 3, 2021.

71. * “Producer Beliefs and Conservation: The Impact of Perceived Water Scarcity on Irrigation Technology Adoption”, NBER: Economic Perspectives on Water Resources, Climate Change, and Agricultural Sustainability, May 12th, 2022 (presenter: Joseph Blumberg, included Dale Manning)

70. * "Conservation and Insurance Policy Interactions: The Direct and Indirect Effects of Groundwater Policy and Implications for the Distribution of Costs", AAEA Annual Meeting 2022, Anaheim, California, August 1st, 2022. (presenter: Calvin Bryan, included: Dale Manning and Matt Sloggy)

69. * Resource Management, Risk, and Insurance: A Decomposition of Groundwater Management Impacts in the Ogallala Region”, AGU Hydrology Days, Fort Collins, Colorado, April 25th, 2022 (presenter: Calvin Bryan, included: Dale Manning and Matt Sloggy)

68. * “Can Investment in Storage Infrastructure Reduce Water Shortages?”, AGU Hydrology Days, Fort Collins, Colorado, April 25th, 2022 (presenter: Ahmed Gharib, included: Mazdak Arabi, Joey Blumberg and Dale Manning)
67. * “When can Additional Storage and Demand Reductions Facilitate Adaptation to Increasing Water Scarcity? Lessons from the South Platte River Basin of Colorado.”, AGU Fall Meeting 2021, New Orleans, Louisiana December, 2021 (presenter: Ahmed Gharib)
66. “An Analysis of the Impact of Drought on Agriculture, Local Economies, Public Health, and Crime Across the Western United States”, NOAA Pacific Northwest December Drought & Climate Outlook Webinar, online, December 13, 2021.
65. “Assessing the Impacts of Recent Drought on Ecosystem Service-based Sectors, Criminal Activity, and Health Outcomes in Arizona, Colorado, New Mexico, and Utah”, W-4190 Regional Meeting, Coeur D'alene, Idaho, October, 14 2021.
64. “Assessing the Impacts of Recent Drought on Ecosystem Service-based Sectors, Criminal Activity, and Health Outcomes in Arizona, Colorado, New Mexico, and Utah”, NOAA Southwest Drought Virtual Forum, online, September 21, 2021.
63. “Assessing the Impacts of Recent Drought on Ecosystem Service-based Sectors, Criminal Activity, and Health Outcomes in Arizona, Colorado, New Mexico, and Utah” NIDIS Regional Drought Taskforce Seminar, online, September 8, 2020
62. “Estimating the Impact of Drought on Ecosystem Service-based Sectors, Criminal Activity, and Health Outcomes in the Western United States” InTERFEWS Seminar Series, Fort Collins, CO, November 1, 2019
61. "Comparative Profitability of Water-Limited Irrigation Strategies under Risk Aversion", 2019 Annual Meeting of the Western Agricultural Economics Association, Coeur d'Alene, ID, United States, June 2019 (presenter: Daniel Mooney)
60. "Profitability of Water-Limited Irrigated Cropping Strategies in Northeastern Colorado: A Stochastic Enterprise Analysis", Hydrology Days 2019, Fort Collins, CO, United States, April 2019 (Presenter: Daniel Mooney)
59. "The impact of irrigated land retirement on groundwater use at nearby wells," Ogallala Water Coordinate Agricultural Project annual meeting, Santa Fe, New Mexico, United States, November 29, 2018 (Presenter: Jordan Suter)
58. "The Interaction of Markets for Temporary and Permanent Water Transfers: An Experimental Analysis," Seminar at the School of Agriculture and Environment, Perth, Western Australia, Australia, March 16, 2018 (presenter: Stephan Kroll)
57. "Demonstration of Hydro-Economic Modelling Framework", Second Annual OWCAP meeting, Manhattan, Kansas, United States, November 7, 2017.
56. "Economic Analysis and Design of Policies to Reduce Colorado's Groundwater Use in the Northern High Plains Ground Water Basin", South Platte Roundtable. Longmont, Colorado, United States, July 11, 2017.
55. ^ “Water Pricing and Valuation”, Public Policy and American Drinking Water, Marquette Law School, September 7th, 2016.

54. "Hydroeconomic modeling framework for assessing vulnerability to water demands in arid regions", 2017 UCOWR/NIWR Annual Conference, Fort Collins, Colorado, United States, June 15, 2017. (presenter: *Andre Dozier)
53. "Heterogeneous water demands: Implications for conservation and welfare", 2017 UCOWR/NIWR Annual Conference, Fort Collins, Colorado, United States, June 14, 2017. (presenter: *Alexander Maas)
52. "Residential water and electricity: Substitutes or complements?", 2017 UCOWR/NIWR Annual Conference, FORT COLLINS, CO, United States, June 14, 2017. (presenter: *Alexander Maas)
51. "Rural to urban water transfers and the future of rural agricultural economies in the semi-arid West", 2017 UCOWR/NIWR Annual Conference, FORT COLLINS, CO, United States, June 14, 2017. (presenter: *Kevin Crofton)
50. "The impact of preference aggregation on support for groundwater management policies", 2017 UCOWR/NIWR Annual Conference, FORT COLLINS, CO, United States, June 14, 2017. (presenter: *Aaron Hrozencik)
49. "Modeling Framework for Assessing Tradeoffs in Agricultural and Municipal Water Use", ASCE World Environmental & Water Resources Congress 2017, Sacramento, California, United States, May 2017. (presenter: *Andre Dozier)
48. "Putting Prices into Context: How do Weather and Policy Related Events Impact Effectiveness of Increases in the Marginal Price of Water", 2016 Annual Summer Conference, Association of Environmental and Resource Economists, June 10th, 2016. (presenter: *Janine Stone)
47. "Groundwater Management Policy Evaluation with a Spatial-Dynamic Hydro-Economic Modelling Framework", Camp Resources, August 8th, 2016. (presenter: *Aaron Hrozencik)
46. "Evaluating groundwater management policies with an integrated hydrologic-economic model of the Republican River Basin", 36th Annual Geophysical Union Hydrology Days, Fort Collins, CO, United States, March 21, 2016. (presenter: *Aaron Hrozencik)
45. "Water Markets in Semi-Arid River Basins in the Face of Growing M&I Demands and Changing Climate", NSF Water Sustainability and Climate Investigator Meeting, March 10th, 2016.
44. "Saving Colorado Agriculture, One Gallon at a Time: A latent class model of heterogeneous preferences for water policies and their impacts." WAEA Annual Meeting, San Francisco, CA, July 2015. (presenter: *Janine Stone)
43. "The Effect of Water Institutions on the Value of Water Storage in the West: Lessons from the Colorado-Big Thompson Project." WAEA Annual Meeting, San Francisco, CA, July 2015. (presenter: *Alexander Mass)
42. "Modeling Imperfectly Competitive Water Markets in the Western US." AAEA Annual Meeting, San Francisco, CA , July 2015. (presenter: *Allie Bauman)
41. "Climate Change and Crop Production in the Western United States: The role of surface water irrigation." WAEA Annual Meeting, San Francisco, CA, July 2015. (presenter: Dale Manning)
40. "Heterogeneous Responses to Utility Water Prices: A latent class approach.", Camp Resources XXI, Center for Environmental and Resource Economic Policy, NC State University, August 2014. (presenter: *Janine Stone)
39. ^ "The Economic Impacts of Drought: Looking back at the droughts of 2011 and 2012", Agricultural, Environmental and Water Law Symposium 2014, CLE, July 24, 2014.

38. "Evaluating Heterogeneity in Response to Utility Water Policies: A latent class approach.", WAEA Annual Meeting, June 2014. (presenter: *Janine Stone)
37. "Modeling Welfare Outcomes of Water Markets with Transaction Costs: An agent-based modelling approach.", WAEA Annual Meeting, June 2014. (presenter: *Allie Bauman)
36. "Quantifying the Relationship Between Irrigation Activities and Wetlands in a Northern Colorado Watershed: Assessing this Added Value of Irrigation." American Water Resources Association 2013 Spring Specialty Conference, March 25, 2013. (presenter: *Meagan Smith)
35. "Resource Economics: The Impact on Water Security", Alliance of Crop Soil and Environmental Societies Annual Meeting, American Society of Agronomy, November 5, 2013.
34. "Closing the Gap: Examining Preferences for Meeting Future Water Demands in Colorado", CU Environmental and Resource Economics Workshop, October 5, 2013. (presenter: *Janine Stone)
33. "Performance and Acceptability of Pigouvian Tolls in a Bottleneck Congestion Game: Evidence from the Lab", CU Environmental and Resource Economics Workshop, October 5, 2013. (presenter: *Nicholas Janusch)
32. "Price Signals from Combined Residential Energy and Water Utility Bills", CU Environmental and Resource Economics Workshop, October 5, 2013. (presenter: *Katherine Liesel Hans)
31. "The Effect of Information on Consumer Behavior: Lessons from a Laboratory Experiment", Western Load Research Conference, September 18, 2013. (presenter: *Katherine Liesel Hans)
30. "Household Water and Energy Use: An experimental analysis of decision-making under uncertainty and penalty rate structures.", Camp Resources 2013, August 4, 2013. (presenter: *Katherine Liesel Hans)
29. "Performance and Acceptability of Pigouvian Tolls in a Bottleneck Congestion Game: Evidence from the Lab", Western Economic Association International, July 2, 2013. (presenter: *Nicholas Janusch)
28. "The Effect of Information on Consumer Demand under Alternative Pricing Schemes: An Experimental Analysis", Western Economic Association International, June 30, 2013. (presenter: *Katherine Liesel Hans)
27. "Economics and Its Impact on World Water Security", AAAS Annual Meeting, February 17, 2013.
26. "Integrating Different Approaches to Analyzing a Drought Event's Impact on Communities: Applying EDM and IO Models to the Analysis of Drought in Southeast Colorado." 2012 AAEA Annual Meeting, Seattle, WA, August 2012. (presenter: *Allison Bauman)
25. "Deciphering Household Understanding (or Lack Thereof) of a Water Bill." 2012 UCOWR/NIWR Conference, Santa Fe, NM, July 2012. (presenter: *Katherine Liesel Hans)
24. "Estimating the Economic and Social Impacts from the Drought in Southern Colorado." 2012 UCOWR/NIWR Conference, Santa Fe, NM, July 2012. (presenter: *Allie Bauman)
23. ^ "2011 Agricultural Economic Impacts of Drought in Southern Colorado." Presented at the 2012 Colorado Water Conservation Board Drought Conference: Building a Drought Resilient Economy through Innovation, Gunnison, CO, invited for September 2012.
22. ^ "Impacts to Basins of Origin." Presented at the Western Governors Association' Innovative Water Transfers Workshop, Western States Water Council & Colorado Water Conservation Board, Denver, CO., October, 2011. (presenter: James Pritchett)

21. ^ “Price versus the Value of Agricultural Water.” Presented at the 2011 Colorado Water Congress, Steamboat Springs, CO, August, 2011.
20. “The Impact of Leasing on Water Rights Markets: An Experimental Analysis.” Presented at the CSU Dept. of Ag and Resource Economics Seminar Series, Fort Collins, CO, March, 2011.
19. “The Impact of Leasing on Water Rights Markets: An Experimental Analysis.” Presented at the University of Wyoming Dept. of Ag and Applied Economics, Laramie, WY, March, 2011. (presenter: Stephan Kroll)
18. “Evolution of Household Water Demand Following Drought: Lessons from a Major Colorado Utility.” Presented at the Annual RM-AWWA Conference, Loveland, CO, September, 2011.
17. “The Impact of Leasing on Water Rights Markets: An Experimental Analysis.” Presented at the 2010 South Platte Forum, Longmont, CO, September, 2011. (presenter: S. Kroll)
16. ^ “Water in the West: Current Use and Future Challenges.” Presented at the 13th Annual Farmers Cooperatives Conference, Broomfield, CO, December 2010.
15. “Feasibility Study to Assess the Potential of Urban Water Conservation to Meet Future Demand.” Presented at the CO WaterWise 2nd Annual Conservation Workshop, Denver, CO, September 2010.
14. ^ “Valuing Water: Taking into consideration the timing, location, and type of use.” Presented at the U. of Nebraska Water Law, Policy and Science Conference, Lincoln, Nebraska, April 2009.
13. ^ “Economic Incentives for Demand Reduction.” Presented at the 2009 CU Natural Resources Law Center Annual Conference: Western Water Law, Policy, and Management, Boulder, Colorado, June 2009.
12. ^ “Allocating Water Among Competing Uses: The Potential for Water Markets in New Zealand.” Presented at the Institute for the Study of Competition and Regulation Seminar Series, Victoria University of Wellington, Wellington, NZ, July 2008.
11. ^ “The Medium-term Impacts of Natural Disasters in Brazil.” Presented at the Williams College Symposium on Global Warming and Developing Countries, Williamstown, MA, April 2008.
10. “The Medium Term Impacts of Natural Disasters in Brazil.” CSU Water Seminar, Fort Collins, CO, April, 2008.
9. “Water Use in the Western U.S.: Household Knowledge, Priorities and Willingness-to-Pay.” Presented at the USCID Fifth International Conference: Irrigation and Drainage for Food, Energy and the Environment. Salt Lake City, UT. November, 2009. (presenter: *Jennifer Thorvaldson)
8. “Residential Water Demand: Lessons from Aurora, Colorado.” Presented at the Climate Predication Application Science Workshop, Chapel Hill, NC, March 2008.
7. Managing Residential Water Demand during Drought in Aurora.” Presented at the VPR Global Water Colloquium, Fort Collins, CO, March 2008. (poster)
6. “Residential Demand Management as a Drought Coping Tool: Lessons from Aurora, Colorado.” Presented at the American Water Resources Association 2007 Annual Conference, Albuquerque, NM, November 2007.
5. “Modeling the Future of Water Resources: An Overview and Results from the South Platte Regional Assessment Tool.” Presented at the South Platte Forum, Longmont, CO, October 2007.
4. “Managing Residential Water Demand during Drought in Aurora.” Poster presented at the South Platte Forum, Longmont, CO, October, 2007.

3. “Quantity Uncertainty: The Case of Water Smart Readers.” Presented at the 2007 University of Colorado Environmental and Resource Economics Workshop, Vail, CO, September 2007.
2. “Managing Residential Water Demand: Lessons from Aurora, CO.” Presented at the Center for Science and Technology Policy Research (CSTPR) seminar series, Boulder, CO, March, 2007.
1. “Behavioral and Econometric Lessons from Urban Water Demand under Extreme Drought: Focus on Aurora, Colorado.” Presented at the DARE & Department of Economics U.S. Forest Service Rocky Mountain Research Station Spring 2007 Seminar Series, Fort Collins, CO, February 2007.

TEACHING

COURSES TAUGHT

+indicates graduate level offering

- +Applied Microeconomic Theory (Colorado State University)
- +Advanced Water Resource Economics (Colorado State University)
- Water Resource Economics (Colorado State University)
- Water Law, Policy, and Institutions (Colorado State University)
- Principles of Microeconomics (Colorado State University)
- Assessing the Food, Energy, Water Nexus (co-taught, Colorado State University)
- Agricultural and Resource Economics (Colorado State University)
- +Business Economics for the World Market (Colorado State University)
- Issues in Environmental Economics (Colorado State University)

GRADUATE STUDENT ADVISING (completed)

PhD

Joey Blumburg, 2024 (PhD)

Dissertation Title: Three essays on the economics of water resources and climate change

Current Position: ORISE Postdoctoral Research Fellow in the Human Dimensions Department of the U.S. Forest Service

Alexander Maas, 2017 (PhD)

Dissertation Title: Human Behavior in the Context of Water Scarcity

Current Position: Assistant Professor, Department of Agricultural Economics and Rural Sociology, University of Idaho

Jenny Apriesnig, 2017 (PhD)

Dissertation Title: A Bioeconomic and general equilibrium framework to address fishery management and invasive species

Current Position: Assistant Professor, College of Business, Michigan Tech

Allison Bauman, 2017 (PhD)

Dissertation Title: Modelling the Impact of Transaction Costs, Conservation and Alternative Supply Sources on Water Market Activity in the Western U.S.

Current Position: Research Associate, Department of Agricultural and Resource Economics, Colorado State University

Janine Stone, 2015 (PhD)

Dissertation Title: Three Essays Exploring Heterogeneity in Water Policy Preferences and Policies

Current Position: Assistant Professor, Department of Economics, California State University at Chico

Katherine Hans, 2014 (PhD)

Dissertation Title: Decision-Making under Uncertainty and Imperfect Information: Theoretical, Experimental and Empirical Analyses of Water and Energy Use

Current Position: Water Conservation Program Manager, Fort Collins Utilities

MS

Daniel Villar, 2015 (MS)

Thesis Title: An Analysis of Housing Values and National Flood Insurance Reform under the Biggert-Waters Act of 2012.

Justin Fields, 2014 (MS)

Thesis Title: Analyzing Consumer Response to Time of Use Electricity Rates.

Larissa Serbina, 2012 (MS)

Thesis Title: Describing and quantifying revenue risk producers face when adopting water conserving cropping systems.

Ron Nelson, 2012 (MS)

Thesis Title: 2012 drought in Colorado: estimates of foregone revenues, indirect and induced economic activity for the crop sector.

Cathay Thomas, 2009 (MS)

Thesis Title: A cost-benefit analysis of preventive management for zebra and quagga mussels in the Colorado-Big Thompson System.

OUTREACH

*indicates graduate student

WATER UTILITY FOCUSED OUTREACH

“Overview: SCWSD Rate Study.” Presentation to Spring Canyon Water and Sanitation District, online, November 17, 2021.

“Understanding Commercial Water Demand.” Presentation to Fort Collins Utilities, Fort Collins, CO, July 31, 2018. (presenter: *Matthew Flyr)

“Trends and Drivers of Commercial Water Use: What we know, don’t know, and why we don’t know it.” Liquid Arts Forum, Fort Collins, CO, March 22nd, 2018.

“Estimating the Effectiveness of Water Conservation Programs in Colorado.” Presentation to the CWCB WCTAG, Denver, CO, May, 2012

“Urban Water Demand and Conservation Research Needs in Colorado: Options and Priorities for Future Research.” Presentation to the CWCB WCTAG, Denver, CO, January 2011.

“Estimating the Cost Effectiveness of Water Conservation Programs.” Presentation at the Colorado Water Institute Quarterly Meeting, Denver, CO, November, 2010.

“Residential Water Demand Management.” Presentation to Aurora Water and Citizens Advisory Board, Aurora CO, May, 2009.

Developed “Water Management and Drought” website. The site is a clearinghouse for information on water demand management, drought response, drought impacts, etc.
(http://www.colorado.edu/water_management_and_drought/)

AGRICULTURAL USER FOCUSED OUTREACH

Colorado Ag Water Summit, panel discussion, Aims Community College, Greeley, CO, December 2, 2025.

“Mirasol, Looking at the Sun”, panel discussion. Northern Water, Berthoud, CO, November 3, 2025.

“Mirasol, Looking at the Sun”, panel discussion. Lincoln Center, Fort Collins, CO, October 13, 2025.

“The Allocation and Reallocation of Water in Colorado and the West: An economist’s perspective.” CSU Ag Education Day at the Colorado Farm Show, Greeley, CO, January 25, 2018.

"The Allocation and Reallocation of Water in Colorado and the West: An economist’s perspective." CSU Ag Education Day at the Colorado Farm Show. Demonstration, Greeley, CO, United States, January 25, 2018.

Several Workshop Presentations to Producers in Republican River Basin, eastern Colorado, 2016.

“Who Wins and Who Loses? Water Markets Under Imperfectly Competitive Conditions.” Upper Colorado River Basin Forum, Grand Junction, CO, October 29, 2015.

“Value of Agricultural Water: Matching Preferences and Policies.” Governor’s Forum on Colorado Agriculture, Denver, CO, February 2015.

“Estimating the Impacts of the 2011 Drought on the Arkansas and Rio Grande River Basins.”, Presentation to CWCB representatives, Fort Collins, CO, April 2012

“FRICO Water Market Experiment Results.” Presentation to CWCB Representatives and Colorado Water Policy Analysts, Boulder, CO, January, 2011.

“Alternatives Water Transfer Methods.” Presentation and round table discussion with sample of S. Platte farmers (Prime Rib Club), Brighton, CO, August, 2009

“An overview of CSU's role in the FRICO and Parker projects.” Presentation at the CWCB Ag Water Transfer Coordination Meeting, Denver, CO, September 2009.

“Alternatives to Permanent Water Transfers.” Presentation at annual FRICO shareholder meeting, Brighton, CO, November, 2009.

“Alternatives to Permanent Water Transfers.” Presentation and round table discussion with sample of Colorado irrigators, Brighton, CO, August, 2009

“Alternatives to Permanent Water Transfers.” Presentation to FRICO Water Board, Brighton, CO, July, 2009.

“Institute for Livestock and the Environment: Water Quality Team.” Poster prepared for ILE workshop, Golden ,CO, December, 2008.

Hosted workshop for members of the Wairarapa Regional Irrigation Trust; a group of irrigators (irrigation district) wanting information on how to evaluate the potential net benefits of constructing a new reservoir and how to create a water market for irrigators in their region. Presented information on cost/benefit analysis, water markets in Colorado and discussed details of how to establish a water market specific to their region, Greytown, New Zealand July, 2008.

“An Overview of Water Resource Issues in Colorado.” Presentation to the Colorado State University Agronomy Club, Fort Collins, CO, October, 2007.

GENERAL OUTREACH

“Understanding the Cost of Water and Options for Pricing” 2024 Poudre River Forum, March 1, 2024, Greeley, Colorado.

”An Overview of Past and Current Academic Research Related to Water Markets”, Bizwest Confluence Colorado Water Summit, July 22, 2021, Loveland, Colorado.

(Panelist) SoGES Water: The Critical Nexus in Colorado’s Future, March 9th, 2022, Fort Collins, Colorado.

“Linking Individual Response to an Integrated Modelling Framework.” presentation on behalf of CSU Office of Engagement to the World Business Council for Sustainable Development, Denver Colorado, March 13, 2019.

"The Economics of Water Resource Management." 2017 Water Fluency, Colorado Foundation for Water Education. Palisade, CO, United States, June 29, 2017.

"Economic Analysis and Design of Policies to Reduce Colorado’s Groundwater Use in the Northern High Plains Ground Water Basin." Colorado Water Congress-POND Meeting. Denver, CO, United States, June 21, 2017.

“Using Economics to Inform the Discussion on Meeting Future Water Demands.” Colorado Water Education Foundation, March 2nd, 2016.

“Rural Reinvented: Exploring Shifts in Rural America.” Fort Collins, CO, Feb 19th, 2016. (Panel Member).

Testimony on HB 15-1038 to the HOUSE COMMITTEE ON AGRICULTURE, LIVESTOCK, & NATURAL RESOURCES, Denver, CO, February 2 2015.

“Overview of the Colorado Water Plan”, Fort Collins Town Forum, Fort Collins, CO, September 26, 2015.

"Real-Time Markets in Water." Towards Dynamic Markets in Electric Power, Water, and Wireless Spectrum, Natural Resource Law Center, University of Colorado. April 23, 2013. (Panel including: Richard Howitt, Mark Squillace, Doug Kenney, and Charles Howe)

“An Overview of Water Markets in the Western United States.” Pakistan Trilateral Watershed Rehabilitation and Irrigation Project, Fort Collins, CO, July 2012.

“The Economic Impact of the 2011 Drought on Southern Colorado.” Presentation to CWCB Board, Gunnison, CO, July, 2012.

“Options for Comparing Model Outputs across Model Runs.” Presentation to Three Lakes Water Quality

Taskforce (includes representatives from EPA, BOR, NCWCD, and USGS), Denver, CO, July, 2011.

“Adaptive Management of Zebra Mussels in Colorado.” Presentation at the Colorado Water Institute Quarterly Board Meeting, Denver, CO, November, 2010.

“Identifying the Benefits and Costs Associated with Various Reuse and Conservation Alternatives.” Presentation to EPA, Washington D.C., July, 2009.

“Review of Afghanistan Diagnostic Assessment and Survey Results.” New Mexico State University Afghanistan AWATT Workshop, Las Cruces, NM, July 2009.

Presenter and trainer, AWATT Diagnostic Assessment and Training workshop, Mazar-e-Sherif, Afghanistan, June 2009.

“South Platte Regional Assessment Tool.” Workshop on Sustainable Water Use Efficiency for Afghanistan, Las Cruces, NM, October, 2008.

“Thoughts on Water Resource Planning in Colorado.” Presentation to the Northern Colorado Economic Development Council: Leadership 2010, Fort Collins, CO, August, 2007.

MEDIA AND OTHER POPULAR PRESS

“How Colorado towns are trying to get some water certainty.” interview for NPR’s The Indicator from Planet Money, May 6th, 2024, <https://www.npr.org/transcripts/1197964453>

“From farms to cities: Analysis shows Colorado-Big Thompson water right ownership changes.” Article quoting me on water markets by Ignacio Calderon in the Coloradoan, March 4th, 2024, <https://www.coloradoan.com/story/news/2024/02/29/water-rights-in-northeast-colorado-data-analysis-shows-use-changes-colorado-big-thompson-water/72747554007/>

“Stewarding the Ogallala Aquifer CSU leads groundswell of research, outreach for combating water crisis”, by Anne Manning in The Source Magazine, Fall 2022 <https://magazine.colostate.edu/stewarding-the-ogallala-aquifer/>

“Confluence: Water rights will transfer, one way or another.” Article quoting me on water markets by Ken Amundson in Bizwest, July 22nd, 2021, <https://bizwest.com/2021/07/22/confluence-water-rights-will-transfer-one-way-or-another/>

“Fort Collins-area water district's rates among the highest in the US.” Article quoting me on local water rates by Jacy Marmaduke in the Coloradoan, February 4th, 2018.

“USDA grants \$4.97 million for research evaluating snowpack and water allocation.” Article outlining snowpack and water allocation research Nicole Shearer in the Nevada Today, September 20, 2018.

“CSU-led Team Investigates Water Depletion in Ogallala Aquifer Region.” Article detailing groundwater work in Republican River Basin with Water Preservation Partnership by Megan Fischer in The Rocky Mountain Collegian, April 25th, 2016.

“CSU Ag and Resource Economists Examine Water Trade-Offs.” Article detailing groundwater work in Republican River Basin with Water Preservation Partnership by John Orr on Coyote Gulch, February 26th, 2016.

“A Price for the Priceless” Interview by Nelson Harvey posted by Water Education Colorado, June 30 2016.

“Does real-time water data really cause people to use less?” Article featuring my research on information and household water use by Chris Austin posted on Maven’s Notebook: California Water, Verbatim, August 18, 2015.

“Real time water meter data causes people to use more water, not less.” Article featuring my research on information and household water use by John Fleck posted on INKSTAIN, August 11, 2015.

“Ogallala Groundwater Study will Set Stage for Colorado Water Conservation Debate.” Article detailing groundwater work in Republican River Basin with Water Preservation Partnership by Brett Walton in Circle of Blue, September 9, 2014.

“The Dynamics of Water Pricing.” Interview posted on popular blog Aguanomics, September, 2011.

“Climate Change Laws Pose Challenges, Opportunities for Agriculture.” Article featuring my research in UA News by Daniel Stolte, April 2010.

“Climate Legislation Likely to Benefit Farmers in Arizona, Colorado, and New Mexico.” Post on popular blog Climate + Energy Project featuring interview, April, 2010.

“USDA gives CSU \$1.2M to Address Water Crises.” Interview in the Rocky Mountain Collegian by Lauren Leete, November, 2009.

“How Consumers Choose Wines.” Article featuring my research in Wines & Vines by Peter Mitham, October 2009.

“Two Scary Pictures.” Article on popular blog Aguanomics featuring my research, April, 2009.

Radio Interview discussing the value of water in agriculture with KTIC- Rural Radio, April, 2009.

“Aurora Study Pinpoints Best Conservation Measures.” Article featuring my research in WaterWise by Kim Frick, Winter, 2008.

“The Price is Right? For Water Conservation, It Better Be.” Interview in Utilimetrics by Betsy Loeff, February, 2008.

“What Prompts People to Save Water.” Article featuring my research in Rocky Mountain News by Jerd Smith, October, 2007.

SERVICE

DEPARTMENT

DARE Online Program Coordinator (2023-present)
DARE Executive Committee (2019-2021)
(Chair) Graduate Program (2018- 2021)
(Chair) Water Center Proposal Review Committee (2016)
(Chair) Department Strategic Plan Development Committee (2017, 2018)
(Chair) Micro Core Exam Committee (2011, 2015, 2016, 2018, 2020, 2023, 2024)
Micro Core Exam Committee (2009, 2010, 2012-2014, 2019, 2021, 2022)
Econometrics Core Exam (2015)
Graduate Curriculum Committee (2015-2017, 2022-present)
Graduate Program Review Committee (2011, 2012)
(Chair) Efficiency Committee (2010)

(Chair) Micro Core Examination Redesign Committee (2009)
Environment and Resource Field Exam Committee (2008, 2009, 2010, 2012)
Strategic Planning Committee member (2008)
Undergraduate Curriculum Committee (2008-2012)

Mentoring Committees:

(Chair) Rebecca Jablonski
(Chair) Nathan Delay
(Chair) Kathie Riley
Jesse Burkhardt
Amanda Leister

Search Committees:

Graduate Program Assistant (2018)
(Chair) Environmental and Resource Economics (2016)
Environmental and Resource Economics (2014)
Livestock Economics (2016)

COLLEGE

Faculty Research Council (2024)

GTA Tuition Taskforce (2016, 2017)

Ram Camp (2007-2012)

Global Water Research Colloquium (GWRC) Faculty Advisory Committee. My contributions included helping organize and develop all day VPR Global Water Colloquium (Title: Global Water: From Conflict to Sustainability Challenges and Opportunities in an Interdependent World), Tuesday, March 25, 2008.

UNIVERSITY

Water Center Executive Committee (2014-present)
Faculty Council Committee on Intercollegiate Athletics (2012-2016)
UGO Grievance Committee (2023)

OFFICE IN PROFESSIONAL SOCIETIES

W2190, chair, vice chair, secretary, and member

PROJECT/GRANT REVIEW TEAMS

NIFA Water Portfolio program review (2017)

Member 2012 Proposal Review Panel for NOAA's Climate and Societal Interactions, Sectoral Applications Research Program

Member USDA-RMA Project Review Team: member of a team of researchers that conducted an extensive review of the USDA-RMA sponsored U. of Nebraska project "Enhancing Irrigation Management Tools and Developing a Decisions Support System for Managing Limited Irrigation Supplies for the High Plains." 2010

